

Le Hoang Dang Khoa

Ho Chi Minh City, Vietnam
(+84) 947 085 416| lehoangdangkhoa2842003@gmail.com

EDUCATION

University of Science

District 5, Ho Chi Minh City

Degree: Faculty of Physics and Engineering Physics

PERSONAL PROJECT

Brainwave Signal Classification

Nov 2021 – Jan 2022

- **Description Objectives:** Classified EEG brainwave signals (delta–gamma) using Discrete Wavelet Transform and MATLAB’s Classification Learner, comparing machine learning models (Naive Bayes, KNN, Ensemble, SVM) based on accuracy and execution time, with applications in EEG denoising, Parkinson’s detection, and biomedical signal analysis.
- **Responsibilities:**
 - Studied theoretical background of EEG signals, brainwave types, and Discrete Wavelet Transform (DWT).
 - Extracted features from EEG signals using DWT for classification purposes.
 - Used MATLAB’s Classification Learner with algorithms such as Naive Bayes, KNN, Ensemble, and SVM to classify and evaluate model performance.
- **Technology:** MATLAB, Discrete Wavelet Transform (DWT), Classification Learner, Machine Learning (Naive Bayes, KNN, SVM, Ensemble)

Simulation of a Traffic Light Circuit Using Digital ICs

8th Dec 2024 – 24th Dec 2024

- **Description Objectives:** Designed and simulated a traffic light circuit using digital ICs such as NE555, 4017, 74LS192, and 4511, consisting of functional blocks for pulse generation, counting, display, and LED control, which was simulated with Proteus and implemented on actual hardware.
- **Responsibilities:**
 - Selected the topic and gathered reference materials related to traffic light circuits and digital ICs (NE555, 4017, 74LS192, 4511).
 - Designed the circuit diagram and divided it into functional blocks: pulse generation, counting, display, and LED control.
 - Simulated the circuit using Proteus to verify logic operation and timing for red – yellow – green lights.
 - Built the actual hardware: tested on breadboard, designed PCB layout, assembled components, and tested the physical circuit.
- **Technology:** Proteus, Digital ICs (NE555, CD4017, 74LS192, CD4511), PCB Layout

SKILLS

Programming language

Python, Java, C++

DBMS

Microsoft SQL Server

Microcontroller Tools

MikroC PRO for PIC, Proteus (for simulation)

Soft Skills

Teamworking, Problem-Solving

Languages

Vietnamese, English (basic communication)

